

# Internship Project Report

**Project Title:** Weather Prediction using Machine Learning **Internship Program:** BharatCares – IBM SkillsBuild

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**Date:** September 2026

## 1. Introduction

The objective of this project is to analyze historical weather data and build predictive models for temperature forecasting.

**Dataset Source:** [ExploreMyData – Weather Dataset](#)

Rows: ~1460 (10 cities × 146 days)

## 2. Data Cleaning

- Missing values handled using imputation.
- Outlier detection results:
  - Precipitation → 62 outliers (6.2%)
  - Wind speed → 6 outliers (0.6%)
  - Cloud cover → 12 outliers (1.2%)
  - Temperature & Humidity → no outliers detected.

Outlier detection results for weather parameters

| Column        | Outlier Count | % Outliers |
|---------------|---------------|------------|
| Precipitation | 62            | 6.2%       |
| Wind Speed    | 6             | 0.6%       |
| Cloud Cover   | 12            | 1.2%       |
| Temperature   | 0             | 0.0%       |
| Humidity      | 0             | 0.0%       |

## 3. Exploratory Data Analysis

- **Descriptive Statistics:**
  - Average Temperature: 16.08°C
  - Average Humidity: 62.6%
  - Average Wind Speed: 15.47 km/h
  - Average Precipitation: 8.4 mm
  - Maximum Temperature: 38.34°C, Minimum: -5.99°C

*Descriptive statistics of weather parameters.*

| Metric | Temperature (°C) | Humidity (%) | Wind Speed (km/h) | Precipitation (mm) | Cloud Cover (%) |
|--------|------------------|--------------|-------------------|--------------------|-----------------|
| Mean   | 16.08            | 62.6         | 15.47             | 8.4                | 27.0            |
| Min    | -5.99            | 38.16        | 0.09              | 0.0                | 0.0             |
| Max    | 38.34            | 87.2         | 42.48             | 73.16              | 79.9            |

- **Monthly Trends:**
  - June–July hottest (27–28°C)
  - December coldest (14.36°C)
  - September–November → heavy rainfall (32–73 mm)
  - Humidity peak in August (83%)

*Monthly average temperature, precipitation, and humidity.*

| Month | Avg Temp (°C) | Avg Precipitation (mm) | Avg Humidity (%) |
|-------|---------------|------------------------|------------------|
| Jan   | 16.56         | 8.08                   | 75.1             |
| Jun   | 27.12         | 11.69                  | 51.3             |

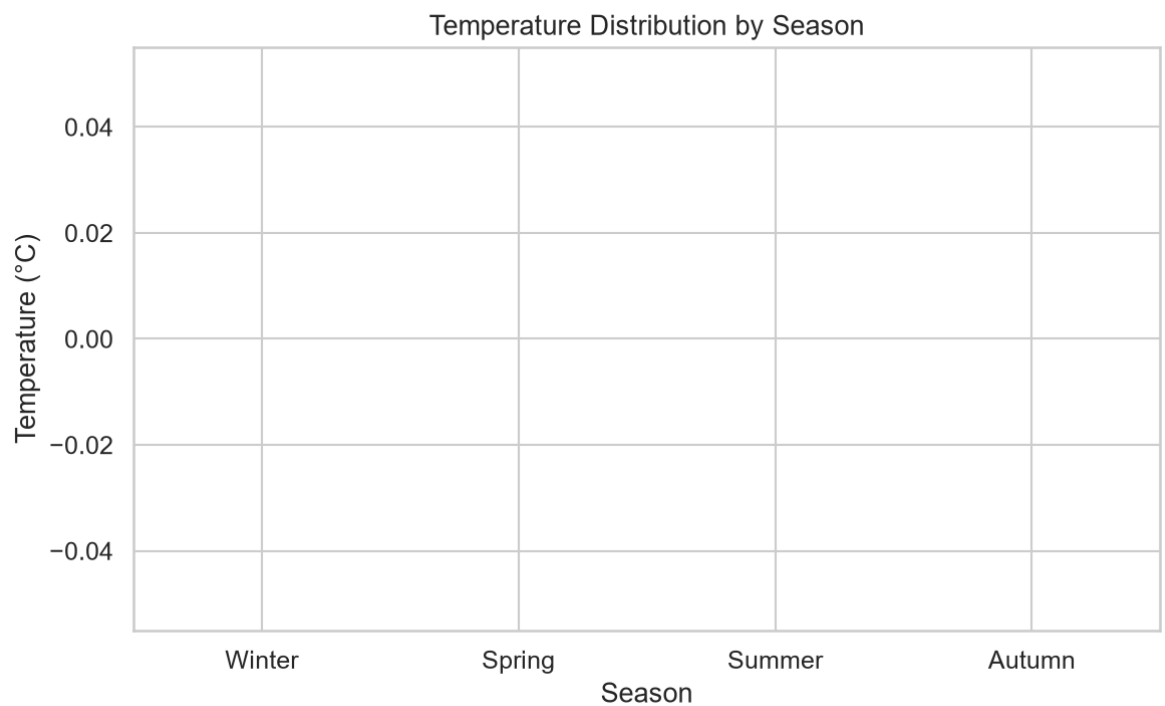
- **Seasonal Statistics:**
  - Summer → hottest (27.26°C, lowest precipitation 4.75 mm)
  - Winter → coldest (3.01°C, humidity 73%)
  - Autumn → moderate temp (9.75°C) but higher rainfall (9.28 mm)
  - Spring → warm (20.79°C) with balanced humidity (58.9%)

Seasonal averages of weather parameters.

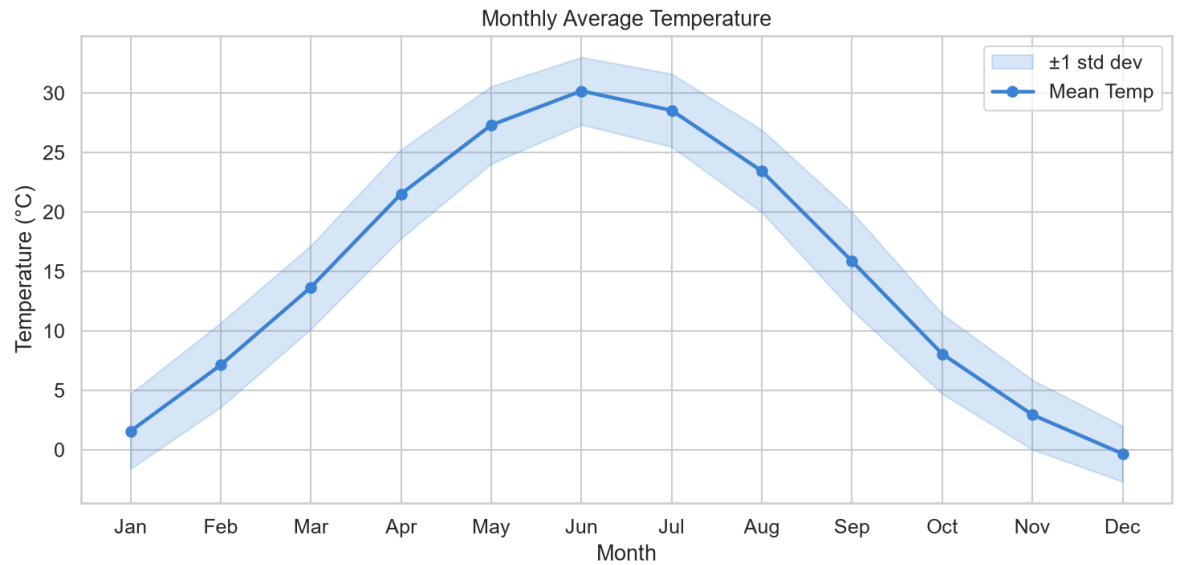
| Season | Avg Temp (°C) | Avg Humidity (%) | Avg Wind Speed (km/h) | Avg Precipitation (mm) |
|--------|---------------|------------------|-----------------------|------------------------|
| Winter | 3.01          | 73.25            | 16.26                 | 7.55                   |
| Summer | 27.26         | 53.34            | 15.1                  | 4.75                   |

#### 4. Visualizations

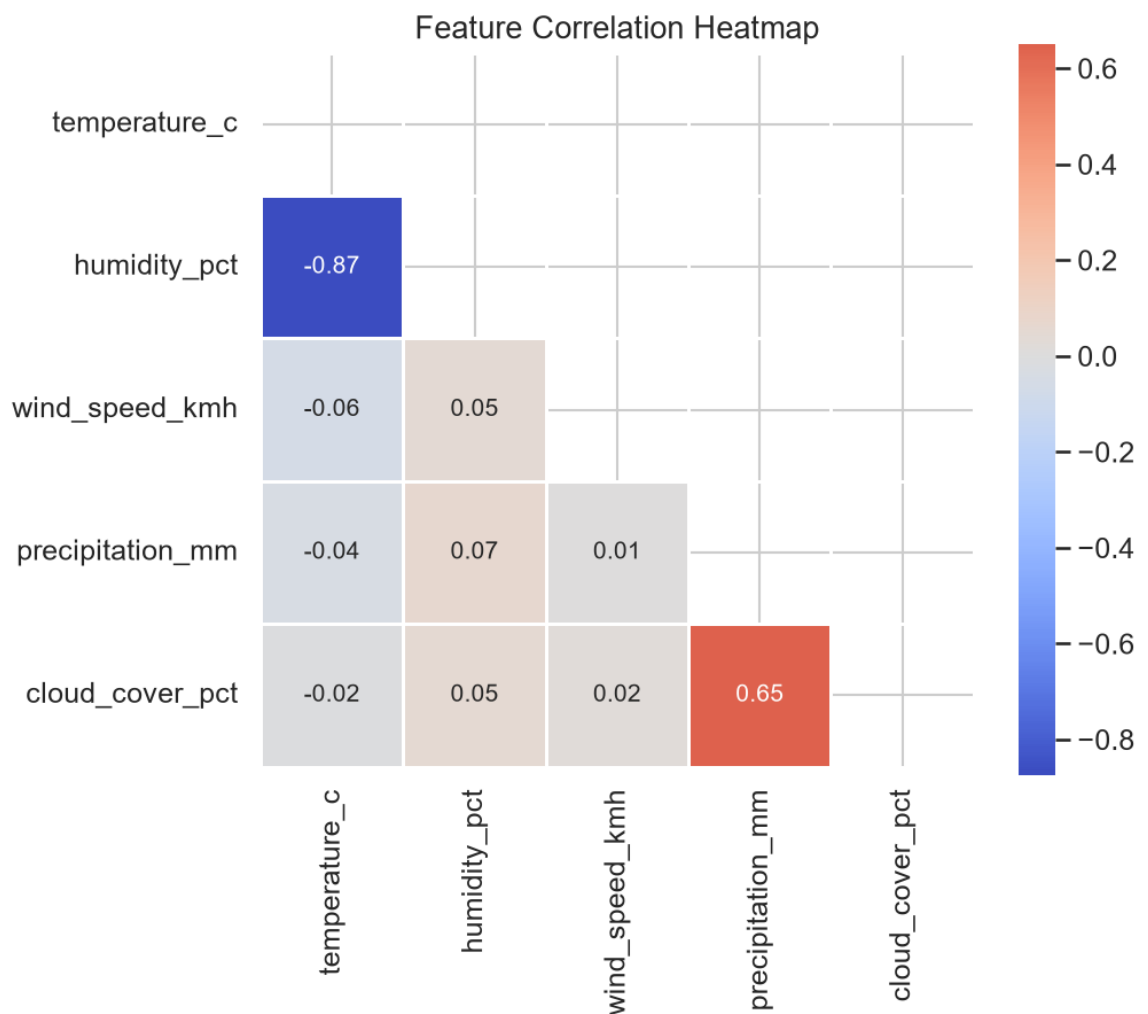
- Seasonal Temperature Boxplot → clear seasonal variation.



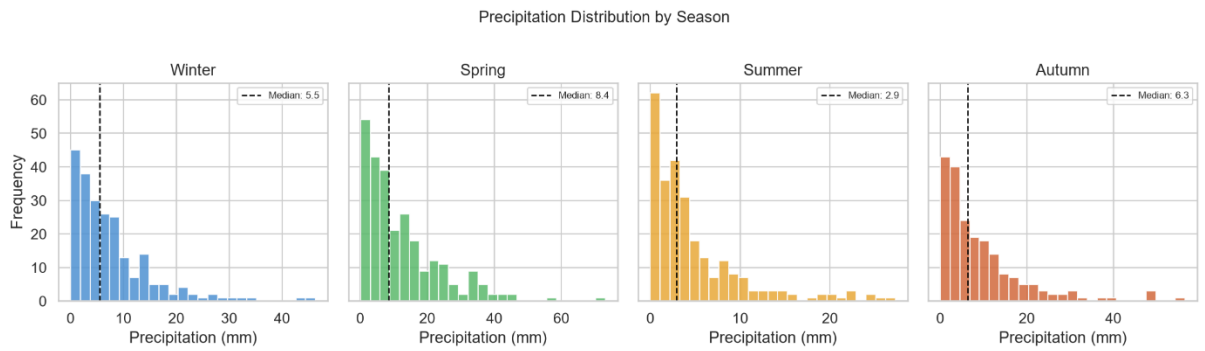
- Monthly Average Temperature Line Chart → peak in June–July, dip in December.



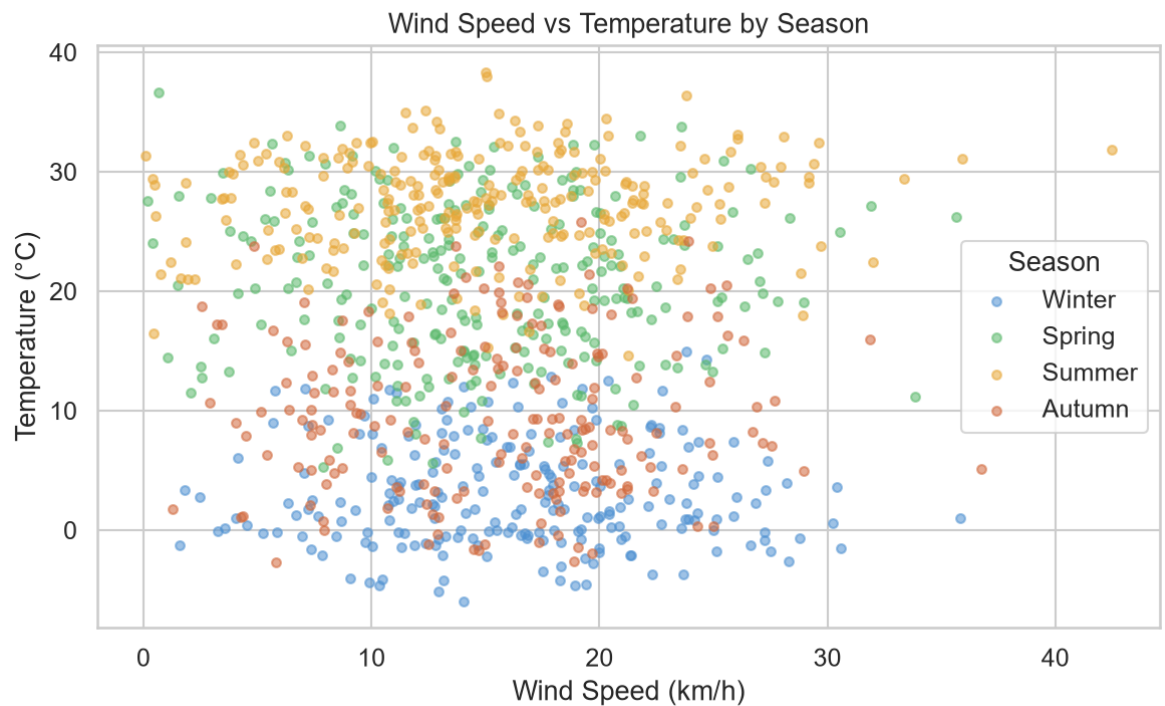
- Correlation Heatmap → strong negative correlation between temperature & humidity (-0.87), positive correlation between precipitation & cloud cover (0.65).



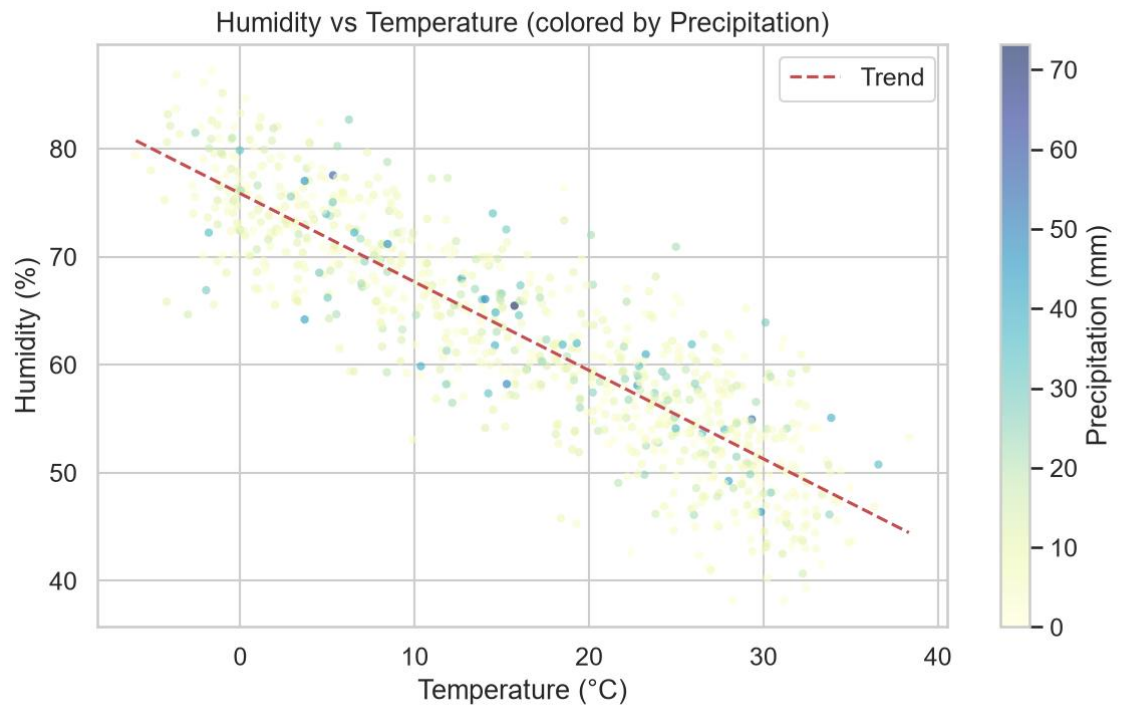
- Precipitation Distribution by Season → summer driest, spring wettest.



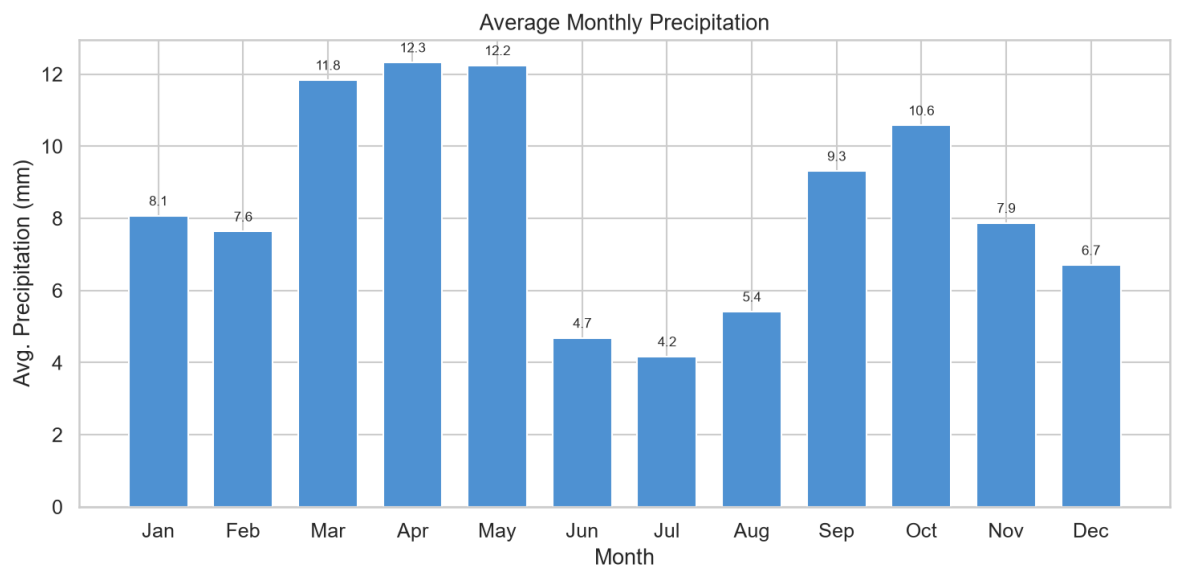
- Wind vs Temperature Scatter Plot → weak correlation.



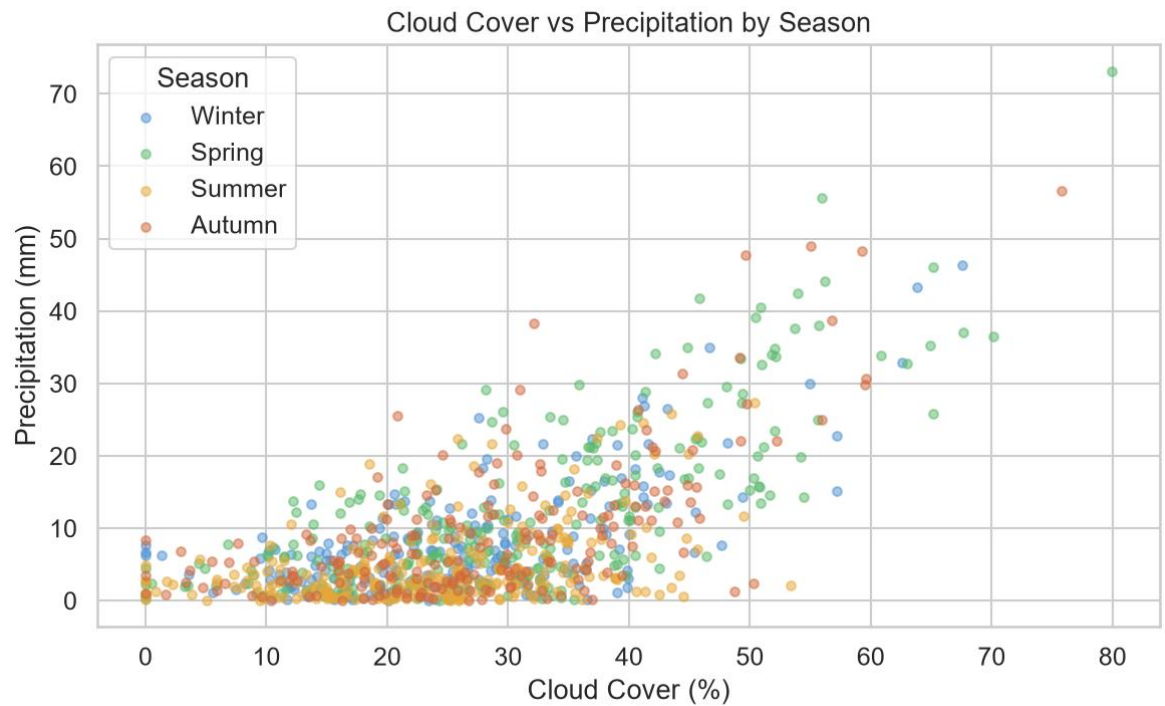
- Humidity vs Temperature Scatter Plot → negative correlation, precipitation influence visible.



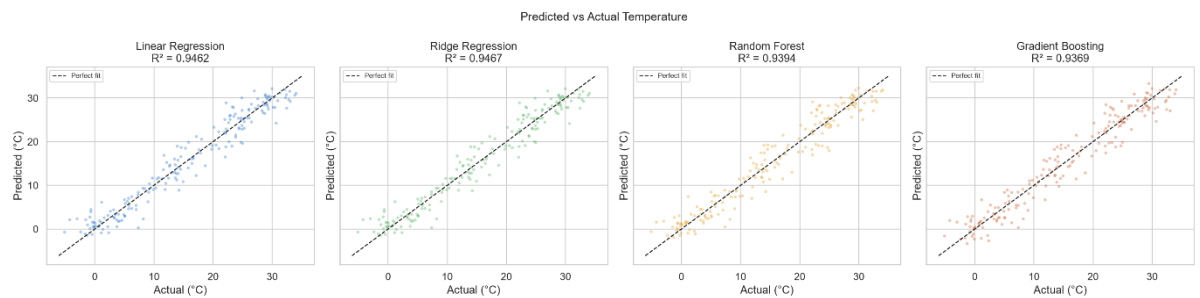
- Monthly Precipitation Bar Chart → rainfall peaks in April–May, drops in June–July.



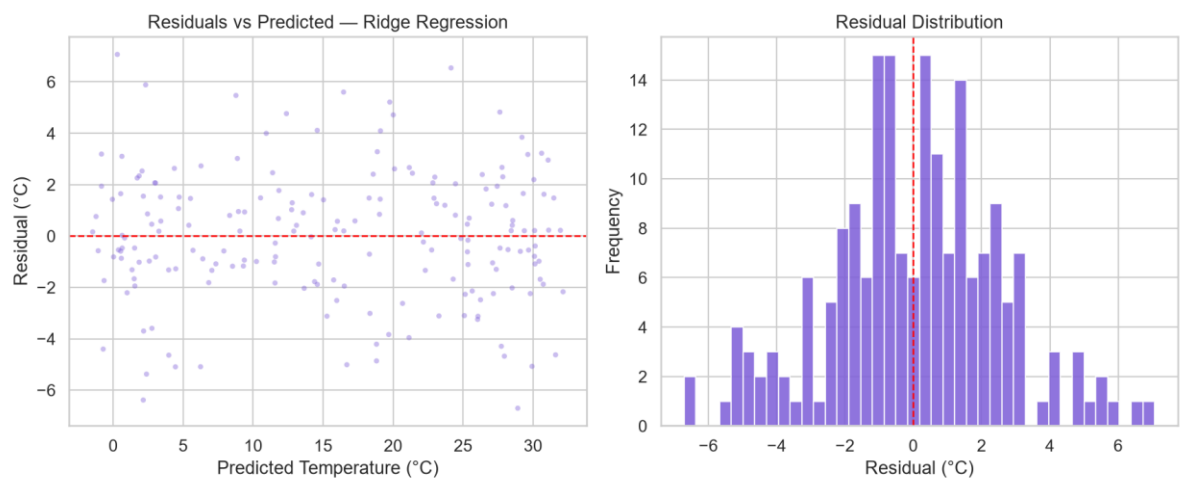
- Cloud Cover vs Precipitation Scatter Plot → higher cloud cover corresponds to higher rainfall.



- Prediction vs Actual Plots → Ridge Regression shows best fit ( $R^2 = 0.947$ ).



- Residuals Plot → errors centered around zero, confirming good model fit.



5. Model Training & Evaluation

- Models tested: Ridge Regression, Linear Regression, Random Forest, Gradient Boosting.
- **Best Model:** Ridge Regression
  - MAE: 1.96
  - RMSE: 2.51
  - $R^2$ : 0.947 (highest among all models).
- Random Forest & Gradient Boosting showed slightly lower performance ( $R^2$  ~0.93–0.94).

*Performance metrics of regression models.*

| Model             | MAE  | RMSE | $R^2$ |
|-------------------|------|------|-------|
| Ridge Regression  | 1.96 | 2.51 | 0.947 |
| Linear Regression | 1.97 | 2.52 | 0.946 |
| Random Forest     | 2.09 | 2.67 | 0.939 |
| Gradient Boosting | 2.17 | 2.73 | 0.937 |

6. Conclusion

- The weather dataset analysis revealed clear seasonal patterns and strong predictive accuracy.
- **Key Insights:**
  - June–July are the hottest months, December the coldest.
  - August shows peak humidity (83%).
  - Ridge Regression provided the most accurate predictions.
- **Future Improvements:**
  - Add more features (pressure, visibility, solar radiation).
  - Experiment with deep learning models for enhanced accuracy.